

Where Classical Interpolation and a Convolutional Surrogate Fail: Wavefunction Super-Resolution on the 1D Quantum Harmonic Oscillator

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Abstract

We train a one-dimensional convolutional network to reconstruct 1024-point ground-state wavefunctions of the quantum harmonic oscillator from 64-point coarse solutions, and compare it with cubic-spline interpolation on 200 matched held-out samples. Both methods clear the $F > 0.99$ fidelity threshold on every sample (network mean 0.999832, spline 0.999790), so the comparison turns on how each method's error is organized over the frequency range. The difference is sharpest where the physics is easiest: in the lowest-frequency bin, where the coarse grid resolves the state best, the spline attains its lowest error while the network attains its highest, on identical inputs. Spline infidelity is perfectly rank-correlated with frequency (Spearman $\rho = +1.000$), a resolution-limited failure, whereas network infidelity follows an inverted-U over the sampled interval, worst at both edges ($\rho = +0.861$ against distance from the midpoint). Retraining on a wider interval moved error off the old edges but not symmetrically onto the new ones, only partly supporting a training-density explanation.

1 Introduction

Direct solution of the time-independent Schrödinger equation (TISE) on a fine grid scales badly: an N -point 1D problem requires diagonalizing an $N \times N$ Hamiltonian, and dimension and particle number compound the cost exponentially. A natural mitigation is to solve cheaply on a coarse grid and reconstruct the fine solution with a learned model. We train a convolutional network, QuantumResNet, to map a 64-point coarse ground state of the quantum harmonic oscillator to its 1024-point counterpart, scored by quantum fidelity.

The 1D harmonic oscillator does not need machine learning; it solves in tens of milliseconds on our hardware. It is a controlled setting with exact ground truth across a densely sampled parameter family. Since a cubic spline clears $F > 0.99$ across the whole range, a higher mean fidelity for the network would say little; the informative question is what organizes each method's error, and the answer differs: the spline's by undersampling, the network's by position within its training range.

Three statements bound the claim. The model is a supervised surrogate, not a physics-informed neural network: its loss is pure mean-squared error against precomputed eigenvectors, with no Schrödinger-residual term, the defining feature of a PINN [1]. “Quantum” here means quantum simulation on classical hardware, with no qubits or circuits. The measured speedup is machine-specific and secondary.

2 Related Work

PINNs embed a PDE residual in the training loss [1]; ours does not. Neural operators such as the FNO learn resolution-independent maps with zero-shot super-resolution [2]; ours is resolution-specific but reconstructs fine fields from coarse observations of a family indexed by ω , as in SRCNN [3]. Closest are convolutional surrogates for the Schrödinger equation: Mills et al. [4] map a 2D potential to a ground-state energy, whereas we reconstruct the eigenvector; Li et al. [5] apply super-resolution-inspired upsampling to electron densities, scoring by energy error. Neural-network quantum states parameterize the wavefunction variationally with no precomputed ground truth [6]. That learned surrogates degrade away from their training distribution is well documented: neural operators fail on out-of-domain PDE inputs at high in-domain accuracy, and Gopakumar et al. [8] give conformal coverage across a surrogate's operating domain under an exchangeability assumption. What this setting adds is separability: a classical baseline runs on identical inputs and its error is ordered by physics without exception ($\rho = +1.000$, zero rank inversions across 200 samples), so the two failure modes can be observed side by side.

3 Methods

System and ground truth. We solve the 1D TISE in atomic units ($\hbar = m = 1$), $-\frac{1}{2}\psi'' + V(x)\psi = E\psi$, with $V(x) = \frac{1}{2}\omega^2 x^2$ on the domain $[-10, 10]$ with Dirichlet boundaries. A three-point finite-difference stencil yields a sparse Hamiltonian; the ground state comes from *eigsh*(H , $k=1$, *which*="SA"), solved independently at $N = 1024$ (target) and $N = 64$ (input). Wavefunctions are L^2 -normalized with a fixed sign convention; only the nodeless $n = 0$ state is used, at fixed potential centre $x_0 = 0$. The state is a Gaussian of width $\sigma = 1/\sqrt{\omega}$ against grid spacing $\Delta x = 20/63 \approx 0.317$, so coarse points per width fall from 4.45 at $\omega = 0.5$ to 1.41 at $\omega = 5.0$.

Data and training. 2,000 samples, $\omega \sim U[0.5, 5.0]$, seed 42; split 72/18/10 (1,440/360/200). QuantumResNet maps $[B, 1, 64] \rightarrow [B, 1, 1024]$ through four transposed-convolution upsample stages (kernel 4, stride 2, padding 1), each followed by a pre-activation residual block (BatchNorm $\rightarrow$ GELU $\rightarrow$ Conv1d, twice, with skip); 64 channels, 166,209 parameters. Adam (lr 10^{-3} , batch 64), MSE loss, cosine annealing over the 30-epoch budget; early stopping (patience 7) restored the best-validation checkpoint. A run takes 113 s on the CPU named in Section 4. Early stopping did not fire, which is not evidence of headroom: the schedule drives the learning rate to near zero by epoch 30 by construction. Robustness checks retrain the identical configuration under seeds 43 to 46, and one single-seed ablation retrains on ω in $[0.3, 7.0]$ with $N = 3,000$ (448 samples per unit ω against the canonical 444; split 2,160/540/300).

Metrics and baselines. Fidelity is $F = (\sum_i \psi_{\text{hr},i} \psi_{\text{pred},i} \Delta x)^2$, both wavefunctions renormalized under the discrete-sum rule of the released code; the analysis works with infidelity $1 - F$. Baselines: a cubic spline (*CubicSpline*, $64 \rightarrow 1024$); Fourier zero-padding applied to the 63 unique periodic points of the endpoint-inclusive grid; and two oracle baselines that assume the harmonic family. The *moment oracle* estimates ω from the coarse input alone through the ground-state identity $\text{var}(x) = 1/(2\omega)$ and emits the analytic ground state $(\omega/\pi)^{1/4} \exp(-\omega x^2/2)$ on the fine grid. The *inversion oracle* instead recovers ω by minimizing $1 - F$ between the input and the coarse finite-difference ground state at ω (bounded Brent, xatol 10^{-8} , seeded from the moment estimate), then emits the fine-grid finite-difference state at the recovered value; it is not a candidate method, costing a fine solve per sample, but a bound on the information the coarse input carries. All share the fidelity function and test slice; none is tuned. Error shapes are summarized by Spearman correlations of infidelity against ω and against $|\omega - \omega_{\text{mid}}|$ ($\omega_{\text{mid}} = 2.750$). The second predictor is a folded transform of the first, so the pair restates Table 1's shapes and does not test between explanations.

4 Results

Fidelity. Train loss fell to 9.755×10^{-6} and validation to 7.895×10^{-6} without divergence. On the 200-sample test set the network reached mean fidelity 0.999832 (median 0.999889, min 0.999328, max 0.999909, std 1.13×10^{-4}), 0/200 below $F = 0.99$; the spline reached 0.999790 (median 0.999831, min 0.999442, max 0.999995, std 1.65×10^{-4}). Across seeds 42 to 46, network mean fidelity is 0.999869 (range 0.999832 to 0.999914) and the spline's 0.999793 (0.999769 to 0.999818); seed 42, the headline run, is the lowest of the five. The Fourier baseline as first implemented scored 0.969008, a grid-convention error rather than a property of Fourier interpolation: the endpoint-inclusive 64-point array was transformed as if periodic in index space, a rigid shift of 0.149 length units at the domain centre. Corrected, it scores 0.999770, comparable to the spline (Table 1). The moment oracle reaches 0.999789 and tracks the spline bin by bin rather than the network, its ω estimate carrying a mean $|\Delta\omega|$ of 0.13. The inversion oracle recovers ω to a mean $|\Delta\omega|$ of 6.09×10^{-8} (max 2.81×10^{-7} , with no optimizer failures) and matches the target to a mean infidelity below 1.1×10^{-16} in every bin, at the level of double-precision round-off. The coarse input therefore determines the state completely within the family, so the losses of the spline and the moment oracle at high ω are those of a family-agnostic estimator rather than missing information, and the network's 1.68×10^{-4} lies far above what the input permits.

Matched comparison. The network wins 132 of 200 matched samples (66%; one-sided binomial $p = 3.5 \times 10^{-6}$; Wilcoxon on paired infidelities $W = 6588$, $p = 2.4 \times 10^{-5}$), with $1.25\times$ lower mean infidelity overall. Both methods clear $F > 0.99$ on every sample, so these are differences in the structure of very small errors, not a pass/fail separation. The aggregate conceals a reversal (Figure 1): the network loses all 36 samples in the lowest bin and 32 of 36 in the second, then wins all 128

above. Below $\omega = 1.5$ it wins 0 of 42; at $\omega \geq 3.5$ it wins 65 of 65 with $2.31\times$ lower mean infidelity. For this seed the spline's last win falls at $\omega = 2.03$ and the network wins every sample from $\omega = 2.09$ up; across five seeds that run begins between $\omega = 1.73$ and 2.10 (mean 1.95) with 122 to 157 wins of 200, so the crossover is a band, not a fixed value.

ω range	n	Spline F	FFT F	Moment F	CNN F	Spline 1-F	CNN 1-F	CNN wins
[0.5, 1.4)	36	0.999980	0.999979	0.999981	0.999680	$2.03e-5$	$3.20e-4$	0 / 36
[1.4, 2.3)	36	0.999932	0.999930	0.999936	0.999894	$6.75e-5$	$1.06e-4$	4 / 36
[2.3, 3.2)	46	0.999836	0.999826	0.999840	0.999896	$1.64e-4$	$1.04e-4$	46 / 46
[3.2, 4.1)	42	0.999708	0.999682	0.999709	0.999892	$2.92e-4$	$1.08e-4$	42 / 42
[4.1, 5.0]	40	0.999527	0.999466	0.999511	0.999777	$4.73e-4$	$2.23e-4$	40 / 40
Overall	200	0.999790	0.999770	0.999789	0.999832	$2.10e-4$	$1.68e-4$	132 / 200

Table 1: Fidelity and infidelity by ω bin; all methods on the same 200 matched samples (seed 42). FFT is the corrected Fourier baseline.

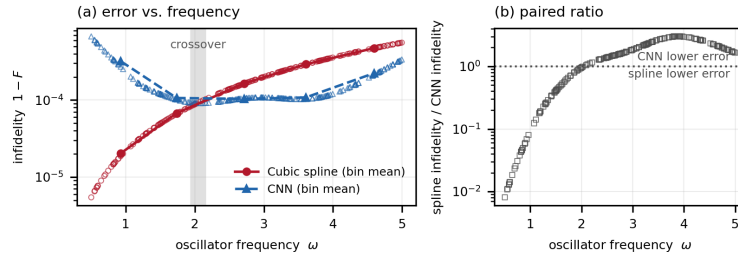


Figure 1: (a) Infidelity against ω for spline and network on the 200 matched samples (open markers) with bin means, log ordinate; the curves cross in the shaded band near $\omega \approx 2$. (b) Per-sample ratio of spline to network infidelity; its crossing of unity locates the same transition.

Two error structures. The clearest evidence comes from where the physics is easiest. In the lowest bin the coarse grid is at its most generous, and the spline attains 2.03×10^{-5} , the lowest error of any method in any bin; on those same samples the network attains 3.20×10^{-4} , its own worst and roughly three times its interior error. Since the spline is at its most accurate exactly there, undersampling cannot be what the network is failing on. Numerically, spline infidelity is perfectly rank-correlated with ω ($\rho = +1.000$, zero inversions) and uncorrelated with the folded predictor ($\rho = +0.031$, $p = 0.66$); network infidelity is uncorrelated with ω ($\rho = +0.026$, $p = 0.71$) and strongly correlated with distance from the midpoint ($\rho = +0.861$, $p = 4.0 \times 10^{-60}$), restating the inverted-U. Re-expressing the predictor as $\sigma = 1/\sqrt{\omega}$ or $\log \omega$ leaves these unchanged up to sign, as monotone reparametrization must, excluding a parametrization artefact without adding evidence. Low-edge infidelity exceeds high-edge by $1.433\times$.

A distributional reading fits this pattern: accuracy set by local training density, worst where a sample has neighbours on one side only. To test it we retrained once on ω in $[0.3, 7.0]$ at matched density. The first prediction held only in part: both former edges improved once interior, but unequally. The former high edge fell from 2.23×10^{-4} to 5.45×10^{-5} , indistinguishable from the interior 5 to 6×10^{-5} , while the former low edge ($[0.5, 1.4)$) improved about threefold, from 3.20×10^{-4} to 1.09×10^{-4} , but remains roughly twice interior. The second did not hold: error concentrated at the new low edge (3.15×10^{-4} for $\omega < 0.5$, $n = 9$) but only mildly at the new high edge (7.40×10^{-5} for ω in $[5, 7]$). Mean fidelity rose to 0.999923 and the crossover moved to $\omega = 1.53$. Edge error therefore follows the training boundary, but asymmetrically, which distance to the boundary alone does not explain; Section 5 offers a candidate reason. The alternative that extremal widths are intrinsically hard for a fixed receptive field is not excluded either. The ablation is one seed, and its new low bin holds nine samples.

Timing (machine-specific). Ten repetitions of the pipeline's 50-run benchmark on an Intel Core Ultra 7 256V (CPU): median 20.6 ms per 1024-point solve against 4.9 ms for the coarse solve plus inference, median speedup $4.1\times$ (range 3.1–4.9). These vary with machine and load and are not evidence of practical acceleration at a tens-of-milliseconds baseline.

5 Discussion

The spline's error is bounded not by the information in its input but by its being family-agnostic: the inversion oracle recovers ω from the same 64 samples to a mean absolute error of 6.09×10^{-8} , so the information is present and a family-blind interpolant simply cannot use it. The network's error is instead a function of training-set design, and the ablation shows that moving a boundary moves the error, though not symmetrically. A candidate explanation is that the relevant density is not in ω but in the shape the network must reproduce: uniform sampling in ω gives a density per unit width σ scaling as $2/\sigma^3$, so broad, low- ω states are sparsest in shape space and stay sparse after widening, while narrow, high- ω states are densely covered. That would produce the observed pattern, but it is an inference, not a tested claim; a direct test would sample uniformly in σ or $\log \omega$. One corollary cuts against the network: a surrogate whose accuracy tracks its training data is weakest exactly where this class of method is meant to be used, where high-resolution ground truth is scarcest. The concern is established [7, 8]; this setting adds isolation, since a physics-ordered classical control on identical inputs makes the network's departure directly observable.

6 Limitations

Scope. Every sample is a harmonic ground state at fixed $x_0 = 0$, with only ω randomized; shifted potentials, other functional forms (double-well and anharmonic are implemented but untrained), and excited states are untested. Test ω values lie inside the training range, so this characterizes degradation toward the training edges, not extrapolation.

Mechanism attribution. The distributional explanation rests on the low- ω asymmetry, the error shape, and one widened-range retrain. That retrain moved edge error inward but left the new edges unequal, so it settles neither the distance-to-boundary account nor the receptive-field alternative, and raises a third candidate, sampling density in shape space, untested. It is single-seed, its new low bin holds nine samples, and it trains on 1.5 times the samples and gradient steps of the canonical run, so its lower interior error is not purely a range effect. The rank correlations add no independent support, and the analysis assumes a one-dimensional parameter space in which "interior" is unambiguous.

Seed variance and configuration. Headline numbers come from one model and one split (seed 42). Across five seeds, network mean fidelity spans 0.999832 to 0.999914 and the crossover spans $\omega = 1.73$ to 2.10, so the crossover values in Section 4 are one realization and the margin varies with seed; the inverted-U with a worse low edge appeared in every seed. The architecture was fixed a priori rather than ablated.

Fidelity is the only accuracy metric. It is an amplitude-weighted global overlap, insensitive to high-frequency content and boundary violation, both visible in this model's residuals. Since the kinetic operator involves a second derivative, a reconstruction scoring well on fidelity may score worse on energy or on $\|H\psi - E\psi\|$; no such metric was computed, and the ranking should not be assumed to survive one.

7 Conclusion

The two methods' errors are organized by different variables, most visibly where the physics is easiest. Spline error is perfectly rank-ordered by ω ; since inverting the coarse eigenproblem recovers ω to 6.09×10^{-8} , the coarse input determines the state within the family, so that failure is representational, not informational. Network error follows an inverted-U that five seeds reproduce. Widening the training range moved error off the old edges but not symmetrically onto the new ones, so training density in the shape the network must produce, rather than in the raw parameter, is the open question. Next steps: a direct test by training uniform in σ or $\log \omega$; physics-based evaluation with energy and residual metrics; and extension to the double-well and anharmonic potentials.

LLM use. LLM assistance was used for drafting, analysis code (including the Fourier correction, the moment and inversion oracles, ablation and seed replication), and auditing of citations and numbers; all results were verified against the released code. **Code and data.** An anonymized snapshot of the code, checkpoints, per-sample metrics and analysis scripts accompanies this submission; the released scripts regenerate every number reported here.

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